7 **(a)** Coil X of 1200 turns of insulated wire is wound on to a soft-iron core.

The coil is connected in series with a battery, a switch and an ammeter, as shown in Fig. 7.1.

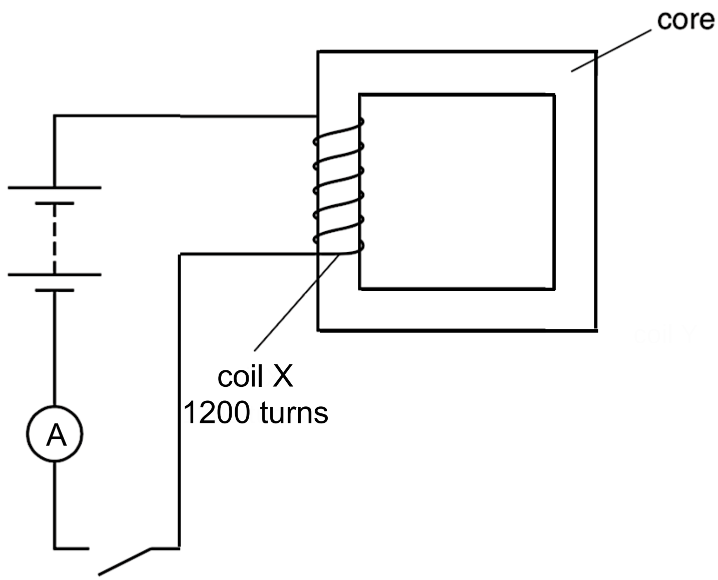

Fig. 7.1

Use the laws of electromagnetic induction to explain why, when the switch is closed, the current increases **gradually** to its maximum value.

....

....

....

....

....

....[3]

(b) Coil Y is now wound on the soft iron, as shown in Fig. 7.2.

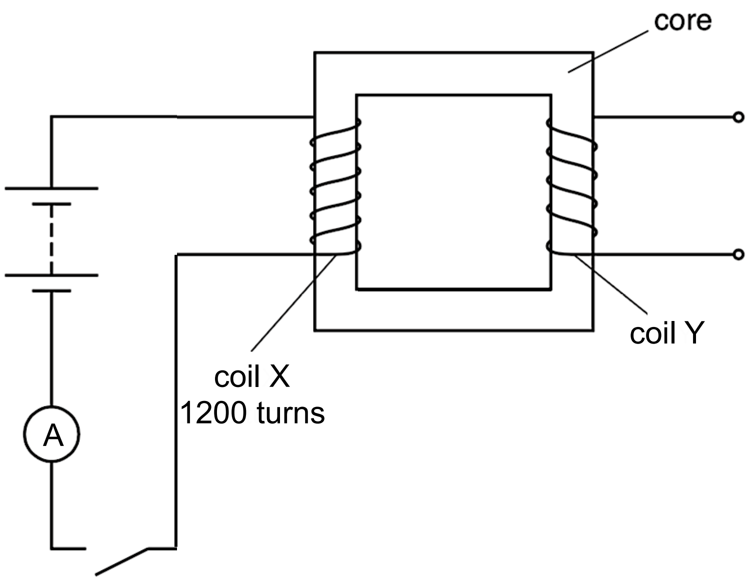

Fig. 7.2

Explain why, when the switch is closed and maximum current is reached in coil X, an electromotive force (e.m.f.) is not induced at coil Y.

....

....

....

....[2]

(c) The battery in **(b)** is now replaced by an alternating voltage of r.m.s. value 90 V. The variation with time *t* of the e.m.f. *E* induced across the secondary coil is shown in Fig. 7.3.

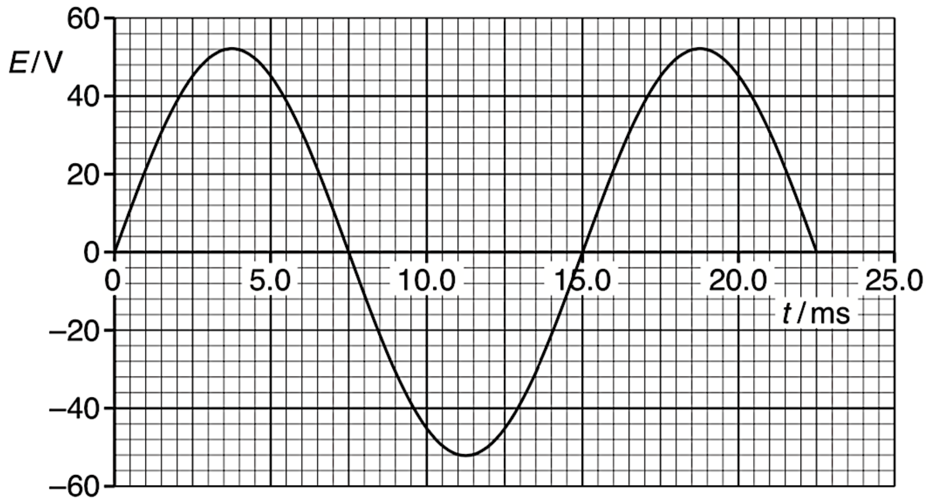

Fig. 7.3

Use data from Fig. 7.3 to

(i) calculate the number of turns of the secondary coil,

number = ... [2]

(ii) state a time when the magnetic flux linking the secondary coil is a maximum.

time = ... ms [1]

[Total: 8]